The effect of offline cognitive processing on online reading behavior in post-stroke aphasia: An eye-tracking study.

Aphasia is a brain disorder that impacts individuals who have suffered an acquired brain injury, characterized by impaired language production and comprehension. Reading difficulties are common in individuals with aphasia, yet they are less understood than speaking impairments (Bullier et al., 2020). Many individuals with aphasia have to read sentences multiple times in order to fully comprehend them. This rereading behavior can reduce an individual's quality of life and can even become maladaptive because excessive rereading is time consuming and may limit reading enjoyment (DeDe, 2017; DeDe & Kelleher, 2021). In a clinical setting it can be hard to determine if individuals with aphasia are engaging in more rereading relative to healthy controls and whether this behavior may be due to aphasia, differences in memory, or both. This project investigates the degree to which verbal memory, visual memory, and aphasia severity influence rereading behavior in English using eye-tracking.

To investigate this, individuals with aphasia (n=20) and neurologically healthy controls (n=16) completed a variety of offline cognitive tasks. A digit span and story recall task assessed verbal memory. A symbol span and a figure copy assessed visual memory. Two coding tasks were included to measure speed of processing. These were accompanied by an eye-tracking-during-reading experiment where participants paced themselves through 176 simple active sentences with 12 words. They then answered yes or no comprehension questions pertaining to each sentence. Sentences were manipulated for semantic constraint, or the degree to which the sentence frame before a specific area of interest (AI) biases the prediction of that word. Half the stimuli were high constraint (e.g., *Before the performance the singer tuned the shiny* ***piano*** *on the stage*). The other half were low constraint (e.g., *In the morning the janitor cleaned the shiny* ***piano*** *on the stage*). We measured the total duration (TD), or how long it took a participant to read the IA (e.g., *piano*), selective rereading duration (SRD), or the time spent rereading previous IA (e.g., *Before the performance the singer tuned the shiny*). Wilcoxon Mann-Whitney U Tests were used to determine that the neurological controls (NC) performed significantly better than *people with aphasia (PWA)* on the forwards digit span, backwards digit span, symbol span, and both decoding tasks. We also used linear mixed effects models to determine whether rereading patterns differ as a function of group, cognitive ability, their interaction, and two random effects (participants and items). Finally, we repeated the same models including PWA and adjusting for aphasia severity.

The Wilcoxon Mann-Whitney U Tests showed that PWA performed worse that NC on forward digit span, backwards digit span, symbol span, a speed of processing task and a fluid intelligence task (all p-values < 0.05). We also found a significant interaction (p-value <0.05) between group and performance on logical memory (both immediate and delayed); individuals with higher logical memory exhibited shorter TD and this effect was stronger in PWA. The models with PWA only showed that individuals with higher speed of processing and higher logical memory (both immediate and delayed) exhibited shorter TD, and individuals with higher performance on symbol span exhibited shorter SRD. These effects were non-significant after adjusting for aphasia severity.

Taken together, these results suggest that rereading behavior in post-stroke aphasia reflects the influence of offline cognitive resources, particularly memory and processing speed, on online reading dynamics. Individuals with stronger memory and faster processing showed more efficient reading dynamics, characterized by shorter total reading times and reduced rereading. However, these relationships were attenuated after accounting for aphasia severity, suggesting that language impairment constrains the extent to which cognitive resources can support online comprehension. Rereading metrics therefore capture interactions between cognitive capacity and aphasia severity, rather than serving as a direct index of language impairment alone.

Table 7.1. MWA Demographic Characteristics

ID	Sex	Hand	Age	Edu	MPO	Race	Ethnicity	WAB AQ	WAB-R Classification
MWA01	F	R	70	22	89	White	Non-Hispanic	97.7	NABW
MWA02	M	L	71	17	101	White	Non-Hispanic	91.2	Anomic
MWA03	F	L	53	20	47	White	Non-Hispanic	92.2	Anomic
MWA04	F	R	63	16	67	White	Non-Hispanic	97.7	NABW
MWA05	F	R	62	18	180	White	Non-Hispanic	96.4	NABW
MWA06	M	R	71	16	85	White	Non-Hispanic	96.3	NABW
MWA07	F	R	54	19	22	White	Non-Hispanic	100	NABW
MWA08	M	R	51	16	56	White	Non-Hispanic	93.8	NABW
MWA09	M	R	67	20	94	White	Non-Hispanic	85.2	Anomic
MWA10	M	R	68	13	72	Black or African American	Non-Hispanic	88.6	Anomic
MWA11	F	L	50	16	199	White	Non-Hispanic	86.9	Anomic
MWA12	M	R	77	15	63	White	Non-Hispanic	73.5	Conduction
MWA13	F	R	61	16	105	Black or African American	Non-Hispanic	86.2	Anomic
MWA14	M	R	73	18	60	White	Non-Hispanic	92.1	Anomic
MWA15	M	R	61	16	565	White	Non-Hispanic	91.3	Anomic
MWA16	F	L	62	16	134	White	Non-Hispanic	96.1	NABW
MWA17	M	R	66	16	44	Black or African American	Non-Hispanic	88.7	Anomic
MWA18	M	R	66	16	87	White	Non-Hispanic	98	NABW
MWA19	M	R	60	16	81	White	Non-Hispanic	63.1	Conduction
MWA20	M	R	54	19	119	White	Non-Hispanic	94.6	NABW
Mean			63	17.0	113.5			90.4	
SD			7.6	2.0	114.7			8.8	

Note. *Edu* = Education (yrs); *MPO* = Months post-onset; *NABW* = Not Aphasic by WAB

Table 1.2 MNC Demographic Characteristics

ID	Sex	Hand	Age	Edu	Race	Ethnicity	MoCA
MNC01	M	R	49	18	White	Non-Hispanic	30
MNC02	M	R	62	16	White	Non-Hispanic	30
MNC03	F	R	66	16	White	Non-Hispanic	28
MNC04	F	R	63	22	White	Non-Hispanic	30
MNC05	F	R	68	17	White	Non-Hispanic	30
MNC06	F	R	68	23	White	Non-Hispanic	28
MNC07	F	R	58	18	White	Non-Hispanic	30
MNC08	M	L	70	18	NR	NR	30
MNC09	M	R	71	27	White	Non-Hispanic	27
MNC10	F	R	70	16	White	Non-Hispanic	28
MNC11	F	R	62	16	White	Non-Hispanic	28
MNC12	M	R	51	19	White	Non-Hispanic	29
MNC13	F	R	73	20	White	Non-Hispanic	26
MNC14	F	R	48	14	White	Non-Hispanic	26
MNC15	F	R	64	18	White	Non-Hispanic	30
MNC16	F	R	47	18	White	Non-Hispanic	29
Mean			61.8	18.5			28.6
SD			8.7	3.2			1.4

References

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